PORTFOLIO · SOFTWARE ENGINEERING | BACKEND & DATA

Tiansawang Suetrong

Digital Science and Technology student at the Faculty of ICT, Mahidol University. I build full-stack systems end to end, from the interface people use to the API and the data model underneath. Two of them are live today.

INSIDE

01 **WasmBot Studio** No-code business rule bots on Rust WebAssembly

02 **HEAT//SHIFT** Map-first urban heat decision platform for Bangkok

03 **More work and contact** Portfolio site, practice projects, skills

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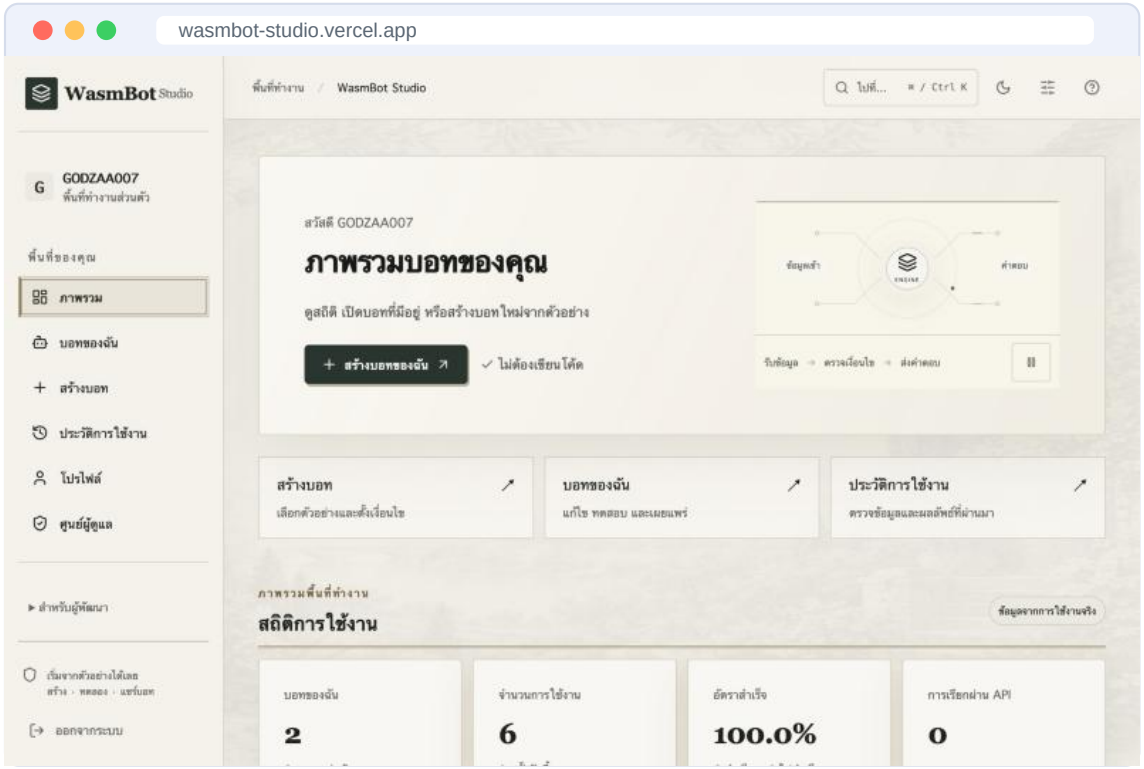


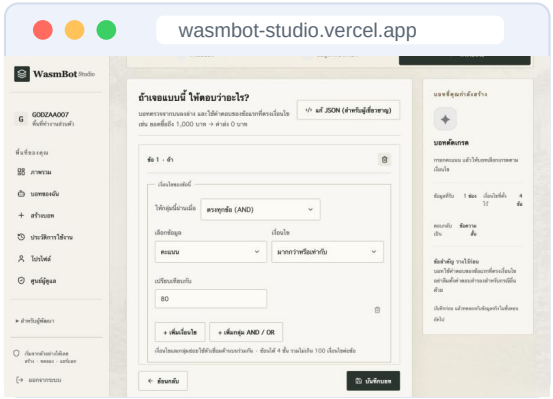
Mahidol University · ICT

B.Sc. Digital Science and Technology · Bangkok

01 / FULL STACK · NO-CODE PLATFORM

WasmBot Studio





Rule editor. Nested AND/OR groups with 12 comparison operators. Both screenshots were captured from the live app on 11 November 2026.

A no-code platform that turns business rules into reusable bots: design them visually, test them on a real Rust WebAssembly engine, publish immutable versions, then use them anywhere.

MY ROLE

Full-stack development: interface, Edge Functions, database, security, and deployment.

WHAT I BUILT

- Visual AND/OR rule editor with **12 operators**, executed by a Rust engine compiled to WebAssembly.
- Immutable, versioned deployments so draft edits never affect the live bot.
- Each bot works as a hosted form, embed, REST API, signed webhook, and typed TypeScript SDK.
- Row-level security, scoped API keys stored as HMAC-SHA256 hashes, rate limits, and audit logs.
- **7 Supabase Edge Functions**, verified with **71 automated tests**.

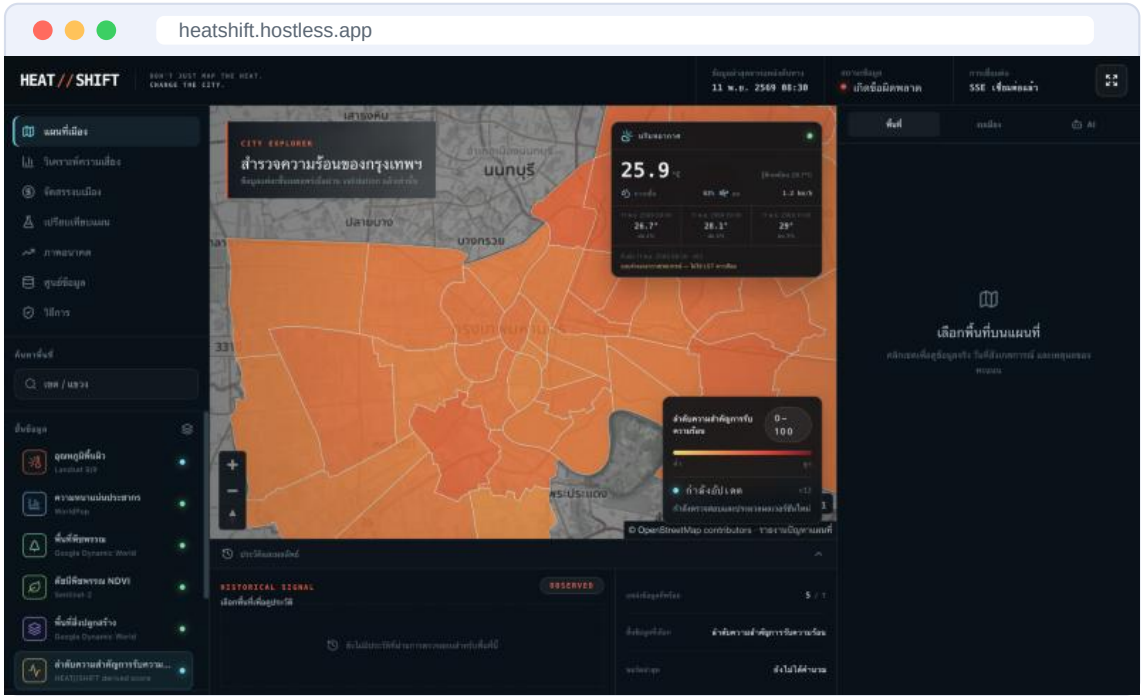
STACK

TypeScript · React · Vite · Tailwind CSS · Supabase · PostgreSQL · Rust · WebAssembly

Live app wasmbot-studio.vercel.app (sign-in required)
Source github.com/tiansawang1818-pixel/wasmbot-studio

02 / GEOSPATIAL · DECISION INTELLIGENCE

HEAT//SHIFT



SEVEN REAL DATA SOURCES, VALIDATED BEFORE PUBLISHING

BMA boundaries 50 Bangkok districts	**Landsat 8/9** Surface temperature	**Sentinel-2** Vegetation (NDVI)	**Dynamic World** Land cover
WorldPop Population	**Open-Meteo** Hourly weather context	**OpenStreetMap** Candidate public sites	**Traceable** Every value versioned

A map-first platform for planning where to invest in cooling Bangkok, built on real satellite, weather, population, and OpenStreetMap data across all 50 districts, with every number traceable to its dataset version.

MY ROLE

System and data workflow design: pipeline, database, planning tools, and cloud deployment.

WHAT I BUILT

- **Six-stage pipeline**: staging, validation, processing, publishing, cache invalidation, live map updates.
- **Seven real sources** update automatically, including Landsat 8/9, Sentinel-2, Dynamic World, WorldPop, and OpenStreetMap.
- Versioned PostGIS datasets; invalid data never replaces the last good version.
- Budget optimizer with Google OR-Tools; forecasts that fail validation are withheld and their errors shown.
- Runs 24/7 on free-tier cloud hosting at **$0 cost**; admin tools behind single sign-on.

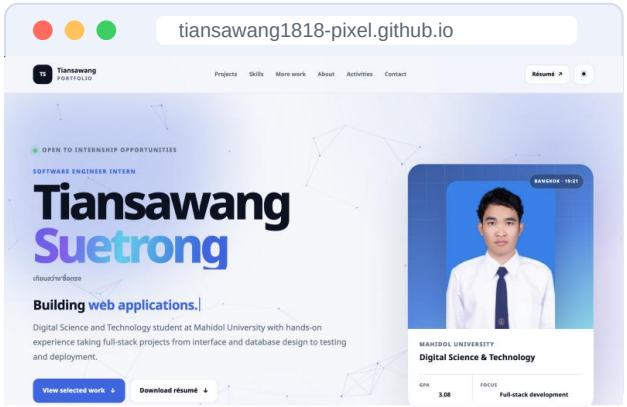
STACK

Python · FastAPI · Next.js · MapLibre · PostgreSQL · PostGIS · Redis · Celery · Earth Engine

Live app heatshift.hostless.app
Source github.com/tiansawang1818-pixel/HEAT

03 / MORE WORK

Smaller builds and practice



This portfolio

HTML · CSS · JavaScript

Buildless, responsive site with dark mode, keyboard navigation, reduced-motion support, and a print layout.

View repository

f(x)

python-calculator

Python

Unit-testing practice: test cases for a calculator class that adds, subtracts, multiplies, and divides.

View repository

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HTML101

HTML · CSS

Web fundamentals labs covering semantic HTML, page layout, styling, and multi-page sites.

View repository

SKILLS

Programming: Python, TypeScript, JavaScript, Java, SQL
Web Development: HTML, CSS, Node.js, Web Services
Databases: PostgreSQL, MySQL, Supabase
Data and Modeling: ER modeling, relational database design, data validation
Tools: Git, GitHub, VS Code

LET'S TALK

Looking for a software engineering, backend, or data internship. Email is the fastest way to reach me.

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